

Focus on Your Focus (FOYF): Cross-Domain Few-Shot Semantic Segmentation by Attention Specialization

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Abstract—Due to data scarcity, privacy, and security, it is difficult to obtain sufficient data to support the research and application of data-driven artificial intelligence (AI) technology across various fields. Although current few-shot semantic segmentation (FSS) methods mitigate performance degradation caused by limited same-domain data, they struggle in cross-domain scenarios with significant domain shifts, limiting their practical utility. Building upon FSS, cross-domain FSS (CD-FSS) generalizes the task to segment novel classes across domains using minimal supervision. However, CD-FSS faces persistent challenges in effective feature pattern representation and transmission across domains. This article comprehensively analyzes existing CD-FSS networks and reveals two crucial insights. First, the inherent inductive bias of the convolutional mode restricts the scene-level contextual information capture. Second, reliance on a single attention inadequately supports complex cross-scene information interaction. To address these issues, we propose an attention specialization network, called “focus on your focus” (FOYF). It integrates multiple attention specialization modules based on Swin Transformer, which profoundly mine intraobject attributes and interobject associations within scenes, dynamically activate cross-scene correspondence links, and hierarchically fuse multigranularity feature patterns. Experimental validation demonstrates the superior performance of FOYF on CD-FSS benchmarks by propagating metalearned scene-level priors, effectively overcoming domain shift bottlenecks. This work establishes a novel technological pathway for cost-effective learning, advancing the development of CD-FSS in data-constrained environments.

Index Terms—Attention specialization, cross-domain few-shot semantic segmentation (CD-FSS), scene-level information, Transformer.

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I. INTRODUCTION

HUMAN beings possess remarkable cognitive abilities that enable rapid recognition of novel concepts with minimal references, even in unfamiliar scenes or tasks. This capability stems from our sophisticated attention allocation mechanisms and cognitive processes, i.e., attention specialization, which allows selective focus on relevant information while filtering out distractions [1]. Specialization refers to the process by which entities develop specialized features through focused differentiation in functions, structures, or objects to adapt to specific needs. In data-constrained domains such as industry, healthcare, and finance, the scarcity, privacy, and security of data pose significant challenges for developing data-driven artificial intelligence (AI) solutions [2]. There is a growing demand for cost-effective learning paradigms that can bridge the gap toward human-level intelligence [3]. To address this, researchers are investigating diverse cost-effective learning strategies, with few-shot learning emerging as a particularly promising approach in the AI landscape.

Cross-domain few-shot semantic segmentation (CD-FSS) represents a critical research direction in few-shot learning, concentrating on developing methods capable of accurately segmenting novel object categories in unseen domains with limited labeled references [4], as shown in Fig. 1. It applies metalearning through episodic training on simulated metatasks across source-target domains within an optimized Siamese network framework. A pair of parallel networks f_θ is trained for extracting the features of labeled support images and query images, separately. These features are then fused to segment the objects. Specifically, it transfers learned priors from a labeled source domain $D_{src} = (I_i, M_i)_{i=1}^{N_{src}}$ to a target domain $D_{tgt} = (I_j, M_j)_{j=1}^{N_{tgt}}$ characterized by significant distribution shifts ($P_{src}(I, M) \neq P_{tgt}(I, M)$) and limited target annotations ($N_{tgt} \ll N_{src}$), where I denotes images and M represents pixel-level annotations. During metatesting, the model receives a support set $S = (I_k^S, M_k^S)_{k=1}^{N_{tgt}}$ containing limited annotated examples from the target domain and a query set $Q = I_{tgt}^Q$ comprising unlabeled target images. The goal is to predict accurate segmentation masks $\tilde{M}_{tgt}^Q$ for all query instances.

CD-FSS has significant academic research challenges and practical application value, making it a cutting-edge research frontier in deep learning fields. Its methods mainly leverage a convolutional mode for feature pattern representation, excelling at capturing fine-grained local object information.

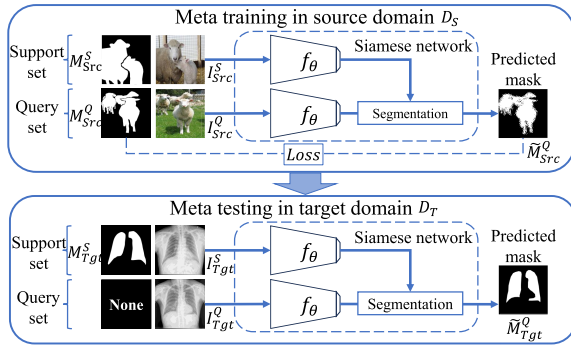


Fig. 1. Schematic of cross-domain few-shot semantic segmentation.

However, they indeed insufficiently consider the interactions between objects, i.e., scene-level contextual information, which is viewed as a key clue for semantic segmentation. Therefore, it is vital for CD-FSS to simultaneously focus on both the intrinsic attributes and external connections of objects. To achieve this goal, we propose a novel network with multiple attention specialization modules based upon the Swin Transformer [5], which can focus on your focus (FOYF). Attention specialization addresses the limited adaptability of general attention across tasks by deeply adapting to the demands of CD-FSS tasks through structural adjustments and modular division. Our approach enables dynamic adaptation to scene-relevant feature patterns while preserving cross-object relational understanding. The main contributions of our work are summarized in the following.

1) We propose the first CD-FSS framework exclusively built upon a Transformer, jointly modeling intrinsic object attributes and external contextual relationships.

2) We design multiple attention specialization modules integrated in the network. The self-attention pattern learning module (SAPLM) captures intrinsic object attributes and external contextual relationships within the scene, the guide-attention pattern learning module activates cross-scene interaction information across scenes, and the attention-based soft fusion module integrates hierarchical scene information within the scene.

3) Experimental results have demonstrated that the Siamese network framework based on attention specialization is promising for CD-FSS.

The remainder of this article is organized as follows. Section II gives a brief introduction to the related work. Section III elaborates on our method FOYF. Section IV reports and analyzes experimental results. Section V is the discussion of FOYF. Finally, Section VI concludes this article.

II. RELATED WORK

A. Semantic Segmentation

Traditional semantic segmentation operates under a closed-set formulation, requiring predefined category spaces where each image pixel is assigned to a predetermined class label for image analysis and task understanding. However, this paradigm inherently lacks open-world learning capabilities, making it fundamentally unable to perform agnostic category

inference. In other words, it cannot generalize to segment novel object categories unseen during the training phase without additional retraining or fine-tuning.

B. Few-Shot Semantic Segmentation

FSS refers to the task of training models to segment novel object classes in images using few labeled support examples per class in the same domain, leveraging prior knowledge acquired from base classes to generalize to unseen categories with minimal supervision [6], [7], [8], [9]. MSGA serves the graph attention map of the query image as prior information to enhance support image segmentation, creating a mutual supervision mechanism. This exploits the correlation between query and support images, enabling better cross-image information utilization and improving query image segmentation [10]. LayerMI maximizes mutual information between support and query features while promoting internal image clustering. This strengthens feature correlations and helps the model better grasp image structures and semantics, leading to more accurate and reasonable segmentation [11]. USD introduces the SAM model for FSS. Leveraging its strong generalization and object-specific extraction capabilities, it adapts well to few-shot scenarios, extracts targeted object features, and shows great potential in enhancing segmentation performance [12]. Metalearning, a paradigm known as “learning to learn,” has significantly advanced FSS capabilities [13]. By constructing numerous episodic tasks, this task-driven strategy enables FSS models to develop category-agnostic understanding, which is critical for generalizing to new classes with limited labels [14]. Existing FSS approaches primarily fall into two categories: prototype learning focuses on class-level feature alignment through statistical representations [15] and affinity learning emphasizes pixel-level correspondence in embedding spaces [16]. Contrastive learning further enhances these methods by optimizing feature discriminability through embedding space manipulation [17]. Although effective for same-domain novel classes, most FSS methods suffer performance degradation under significant domain shifts. This limitation is attributed to three key factors: overreliance on foreground-only feature patterns, inadequate scene-level contextual modeling, and neglect of foreground–background interactions [16], [18]. HSNet [19] exemplifies these issues. Although it uses multiscale dense matching and 4-D convolutions for foreground correlations, its design still prioritizes foreground patterns, lacks scene-level context, and neglects foreground–background dynamics. Thus, even advanced methods such as HSNet remain constrained by these foundational limitations. The inherent inductive biases of convolutional operations, such as locality and translation invariance, restrict efficient modeling of global dependencies and spatial positional associations, thereby impairing models’ ability to understand and reason about long-range semantic relationships across scene regions. To avoid taking out of scene-level contextual information, this article employs attention mechanisms to capture intraforeground, intrabackground, and interforeground–background interactions as much as possible.

The attention mechanism has been proven to have strong capture ability for contextual information and has been suc-

cessfully applied to FSS. With the help of Transformer multiregion aggregation, TPSN can implicitly obtain multiple prototypes and completely replace the average operation, which makes the obtained prototype sensitive to the image context [20]. LATNet proposes a location-aware Transformer module to establish a spatial alignment on the anatomical structures of query and support images through attention to enhance the distinguishability of features and the representativeness of category prototypes [21]. CATrans effectively combines context and affinity measures between support and query using a relation-oriented context Transformer and a relation-oriented affinity Transformer for query segmentation. The Transformer-based cross-attention module is used to strengthen the fusion of support and query convolution features of medical data [22]. Different from previous practices that either adopt semantic-level prototypes from support images or only use foreground support features to encode query features, CyCTR utilizes all pixel-level support features and can effectively eliminate aggregating confusing and harmful support features with the proposed novel cycle-consistency attention to improve segmentation accuracy [23]. SCTrans employs self- and cross-align Transformer modules to boost the feature representations and mine more precise correlations between paired support and query samples [24]. To address the intraclass variation issue, CWTNet introduces a novel metalearning algorithm that leverages a classifier weight transformer for adapting the classifier weights dynamically to each query sample [25]. However, these methods typically integrate Transformer modules as plugins to convolutional frameworks, limiting their ability to model long-range dependencies and scene-level context. Most approaches either use semantic prototypes while ignoring spatial relationships, focus on foreground features without background context, or fail to capture intraclass/interclass interactions. To address these issues, we design self- and guide-attention pattern learning modules based on the Swin Transformer. They serve as unified context extractors, simultaneously capturing scene-level statistical prototypes, dense pixel-level correspondences, and scene-level foreground-background interactions.

C. Cross-Domain FSS

CD-FSS operates under the assumption of consistent data distribution between base (source) and novel (target) domains. However, this assumption often fails in real-world scenarios due to significant domain shifts, leading to dramatic performance degradation. Numerous approaches have been proposed to address this challenge. MMTNet proposes a metaknowledge module to transfer prior source distributions to the target domain [26]. On the contrary, PFMNet designs a prototype-based feature mapping module to transfer high-level features from the target domain to the source domain, similar to a decoder that makes it easier to understand the features [3]. Due to domain shift and extreme imbalance between source and target data, the trained model tends to be dominated by source data, leading to source domain bias and unsatisfactory segmentation effect in the target domain. PATNet, therefore, designs a pyramid-rivet transformation module to

convert domain-specific features into domain-agnostic features, enabling downstream models to better adapt to new domains. Ultimately, segmentation is performed by matching the domain-agnostic features of the support and query sets [4]. Both PATNet and APSEg utilize support prototypes to compute transformation matrices that facilitate the shift from domain-specific features to domain-agnostic features [27]. Due to intraclass variability, support prototypes may not adequately represent the complete information of a single class. The proposal of utilizing query prototypes to augment the set of support prototypes has emerged, aiming to achieve a more effective feature transformation. While feature transformation has been proven to be a potent solution for tackling the CD-FSS challenge, depending exclusively on a sparse set of support images for this transformation is prone to causing overfitting. To address this issue, DMTNet establishes a network based on dual-matching transformation, which simultaneously considers the transformed features of both query and support images [28]. IFA designs a bidirectional few-shot prediction module to establish support-query correspondence [29]. Expand this module to an iterative few-shot adapter within a recursive framework, leveraging iterative support-query correspondence to fully exploit the supervisory signals from limited samples. It is an effective fine-tuning strategy for CD-FSS. ABCDFSS abandons the paradigm of generalizing from the source domain to the target domain and instead learns the segmentation on the target domain in a domain-adaptive manner [30]. Inaccurate feature mapping across domains via the projection layer frequently leads to a failure to preserve the alignment of features that were originally aligned in the original space when projected into the new space. RestNet designs an intradomain residual enhancement module to preserve the matching information of the original feature space in the new space [31]. Compared to domain generalization and domain adaptation, CD-FSS faces even greater challenges in that not only is the target domain different from the source domain seen during training, but the task itself is also novel. Besides, most methods adapt features in a vectorial manner, which fails to learn scene-level semantic context, which is often considered crucial for semantic segmentation [30]. Existing CD-FSS methods focus on cross-domain feature transformation, but this may lead to the loss of critical intradomain information. Building upon these insights, we establish a Siamese network framework with attention specialization optimized through extensive metatask training to capture scene-level contextual information and enhance domain adaptation capabilities.

III. METHODOLOGY

Fig. 2 illustrates the architecture of our model FOYF in the one-shot setting. The model comprises four modules: self-attention pattern learning module (SAPLM), guide-attention pattern learning module (GAPLM), attention-based soft-fusion module (ASFPM), and multiscale attention perception classification module. Given the support image $I^S \in \mathbb{R}^{H \times W \times 3}$, corresponding support masks $M^S \in \mathbb{R}^{H \times W \times 1}$, and query image $I^Q \in \mathbb{R}^{H \times W \times 3}$, where H and W represent the height and width of image, respectively. First, the support image I^S and the

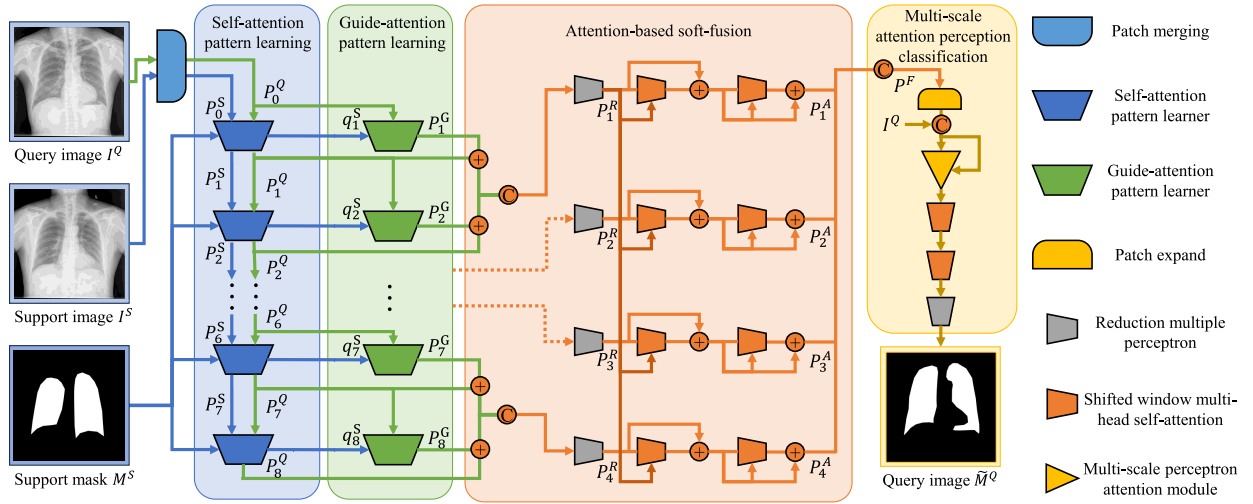


Fig. 2. Architecture of proposed model FOYF.

query image I^Q are input into the SAPLM to obtain the self-correlation pattern at each level. Second, the self-correlation patterns of support and query at the same level are fed into the GAPLM to generate a guide-correlation pattern under the condition of support mask M^S . Third, these self- and guide-correlation patterns of query are fused sequentially at different levels and stages by ASF. Finally, the fused pattern is entered into the multiscale attention perception classification module to predict a pixelwise mask $\hat{M}^Q$. The details are given as follows.

A. Self-Attention Pattern Learning

Current deep learning methods for semantic segmentation primarily rely on convolution or attention mechanisms to capture multilevel feature patterns through hierarchical architectures. As we all know, deep-level feature patterns play an important role in semantic segmentation, while CANet claimed that these patterns may lead to model performance degradation in FSS [32]. This phenomenon is attributed to convolutional neural networks' inductive biases (e.g., local receptive fields), which fail to generalize across domain shifts when deep features dominate. Conversely, attention-based architectures inherently model long-range dependencies, making deeper feature patterns containing scene-level contextual information to enhance performance in both FSS and CD-FSS, as validated by our ablation studies.

Motivated by these findings, we design SAPLM, which fully exploits attention-based shallow and deep feature patterns to address CD-FSS challenges. To accelerate feature pattern learning, this module adopts the pretrained Swin Transformer as its backbone. Unlike Vision Transformers [33] or CNNs, Swin Transformer combines hierarchical architecture with shifted window attention, enabling efficient learning of global-local feature interactions across spatial and depth dimensions.

Leveraging these properties, we design a dual-pathway self-attention pattern learner (SAPL) in the module to model self-correlation patterns for support and query

images separately. The support pathway progressively refines foreground-background interactions. At each $SAPL_i$ block, the previous support pattern p_{i-1}^S derived from the previous $SAPL_{i-1}$ block is elementwise multiplied by an interpolated support mask M^S to focus on foreground regions. The masked feature is then processed by $SAPL_i$ to generate a self-correlation pattern q_i^S

$$q_i^S = SAPL_i(p_{i-1}^S, M^S) = WP(LN(p_{i-1}^S) \odot \phi(M^S)) W_q^S \quad (1)$$

where $\odot$ denotes the elementwise multiplication, ϕ interpolates M^S to match p_{i-1}^S 's resolution, LN is layer normalization, WP represents shift and window partition, and W_q^S is a learnable query matrix of query in the attention mechanism of the Swin Transformer block. Different from the masked support image as input (directly removing background information at the beginning), (1) first mines the intrinsic relationship between the foreground and background, and then filters out the mask background part. The result q_i^S , a prototype-like feature pattern, contains abundant scene-level contextual information, rather than just foreground information.

The query pathway follows the Swin Transformer block to model self-attention among query pattern features. At each $SAPL_i$ block, the input query pattern p_{i-1}^Q coming from the previous $SAPL_i$ is processed as

$$p_i^Q = SAPL_i(p_{i-1}^Q) = MLP(LN(T)) + T \quad (2)$$

where $T = SW_MSA(LN(p_{i-1}^Q)) + p_{i-1}^Q$ and MLP , LN , and SW_MSA , respectively, represent multilayer perceptron, layer normalization, and shifted window multihead self-attention.

The SAPLM inherits the advantages of Swin Transformer by dividing the image into nonoverlapping windows. It models within local windows to focus on the internal attributes of objects. Through the shifted window mechanism, adjacent windows exchange partial regional information at the next layer, establishing cross-window connections. This enables the capture of contextual relationships between objects, effectively balancing the extraction of intrinsic object properties

(e.g., details and textures) and the modeling of external scene-level interactions. By integrating these two pathways, the SAPLM effectively leverages attention-based feature patterns to improve CD-FSS performance, balancing scene-level local detail and global context without convolutional inductive biases.

B. Guide-Attention Pattern Learning

Existing prototype-based methods for FSS typically generate single or multiple prototypes from support images using nonparametric masked average pooling. These prototypes are then compared to query features via distance metrics (e.g., Euclidean or cosine similarity) to predict segmentations. However, representing entire objects with a limited number of prototypes inevitably discards contextual details, leading to suboptimal guidance and inaccurate predictions. While 4-D convolution-based approaches mitigate this by modeling dense pairwise relationships between support and query features, they remain constrained by local feature associations and lack global context awareness.

To address these limitations, we propose a parametric-optimization attention mechanism that dynamically allocates local and global attention to capture cross-scene relationships. Unlike convolutional methods, our approach preserves detailed semantics while integrating hierarchical context. Building on this, we design the guide-attention pattern learner (GAPL), an upgraded Swin Transformer block that aggregates mutual dependencies between support and query features.

Each GAPL_i block takes two inputs: support self-correlation pattern q_i^S produced by SAPL_i at the same level as GAPL_i and query self-correlation pattern P_{i-1}^Q coming from the previous SAPL_{i-1} . GAPL_i then computes a query guide-correlation pattern P_i^G by guiding P_{i-1}^Q with q_i^S

$$P_i^G = \text{GAPL}_i(q_i^S, P_{i-1}^Q) = \text{MLP}(\text{LN}(T)) + T \quad (3)$$

where $T = \text{SW_MSA}|_{q=q_i^S}(\text{LN}(P_{i-1}^Q)) + P_{i-1}^Q$. Here, $\text{SW_MSA}|_{q=q_i^S}$ denotes shifted window multihead self-attention with q_i^S as the query.

By integrating parametric attention in hierarchical GAPL blocks, our method effectively balances scene-level detail and semantic relationships.

C. Attention-Based Soft-Fusion

To address the challenge of integrating feature patterns across varying abstraction levels and spatial resolutions, we propose a hierarchical ASFM. This module dynamically fuses query and support-guided features to preserve fine-grained details while capturing semantic context in the scene, thereby enhancing model robustness in CD-FSS. Specifically, the query self-correlation pattern P_i^Q and the query guide-correlation pattern P_i^G at the same level are combined together via elementwise addition and then performed channelwise max-min normalization to emphasize the scene-correlation feature pattern, so the normalized correlation pattern P_i^N is obtained

$$P_i^N = \frac{P - \min_c(P)}{\max_c(P) - \min_c(P)} \quad (4)$$

where $P = P_i^Q + P_i^G$ and c represents the channel. Because the query guide-correlation pattern contains the scene information of interest that is relevant to the support, it can be used to activate the information of interest in the query self-correlation pattern. Besides, channelwise max-min normalization is applied to maximize and standardize the attention weight of matched information in each channel. This step highlights matched scene-level information between support and query feature patterns while suppressing noise.

Next, the normalized correlation patterns P_i^N from stage j in Swin Transformer's hierarchical structure are concatenated to facilitate information interactions. A learnable linear layer then projects the concatenated features to a unified channel dimension

$$P_j^R = W^R \otimes \text{concat}_j(P_j^N) \quad (5)$$

where P_j^N represents all normalized correlation patterns belonging to the j th stage and W^R is a learnable weight matrix; "concat" denotes the feature concatenation operation. This ensures consistent feature dimensions across stages.

To bridge semantic gaps between stages, we design a skip-attention strategy that activates information from shallower layers with common details to deeper ones with scene-level semantics. It is motivated by the need to balance transferability from shallow features and adaptability to deep features under domain shift and limited data [34]. It ensures that the model leverages the domain-invariant low-level cues to guide the adaptation of domain-specific high-level semantics, resulting in more robust and precise segmentation in the target domain. This rationale aligns with the core challenges of CD-FSS: bridging the domain gap with few samples while preserving spatial precision and semantic coherence. Then, the activated feature patterns are obtained

$$P_j^A = \text{UP}(\text{SW_MSA}(T) + T) \quad (6)$$

where

$$\begin{cases} T = \text{SW_MSA}(q, k, v) + P_j^R \\ k = \text{WP}(P_j^R) W_k \\ v = \text{WP}(P_j^R) W_v \\ q = \begin{cases} \text{WP}(P_j^R) W_q, & j = 1 \\ \text{WP}(\text{PM}(P_{j-1}^R)) W_q, & j > 1 \end{cases} \end{cases} \quad (7)$$

where UP, WP, and PM, respectively, represent the upsampling operation to the same size of P_1^R , shift and window partition, and patch merging operation. W_q , W_k , and W_v are the learnable weight matrixes of query q , key k , and value v in SW_MSA.

D. Multiscale Attention Perception Classification

The activated feature patterns P_j^A from all stages are first concatenated along the channel dimension. To align spatial resolution with the query image I^Q and preserve high-frequency details, we apply a patch expansion (PE) operation. This PE upsamples the concatenated features while reducing channel dimensionality, ensuring spatial fidelity, and enabling fine-grained detail recovery. The upsampled feature map is then concatenated with the original query image I^Q after spatial alignment to form the input for the multiscale attention perception (MAP). There are three parallel branches in the module,

and each branch has two SW_MSA layers and one residual MLP [35] connected in series. Inspired by Yi et al. [36], the window sizes in SW_MSA layers in each branch are, respectively, set to 3, 5, and 7 to make the restoration of detailed context information more complete. Also, each branch includes a shortcut connection, i.e., a residual link for stable training. The outputs of the three branches are concatenated and fused to generate a multiperception feature pattern.

Finally, the fused features pass through the classification head (CH), which consists of two sequential SW_MSA layers followed by a residual MLP. This head refines the features and produces the pixelwise prediction $\tilde{M}^Q$ for the query image. The entire process is summarized as

$$\tilde{M}^Q = \text{CH}(\text{MAP}(P^F)) \quad (8)$$

where

$$P^F = \text{concat}(\text{PE}(\text{concat}_j(P_j^A)), I^Q). \quad (9)$$

IV. EXPERIMENTS

A. Evaluation Setup

1) *Datasets*: We train the model on the PASCAL VOC 2012 dataset with SBD augmentation to establish a robust source-domain representation. For cross-domain validation and testing, we utilize four benchmarks: FSS-1000, Chest X-ray, ISIC2018, and DeepGlobe, following the evaluation strategy proposed in PATNet [4]. This setup ensures a rigorous assessment of model generalization under domain shifts.

2) *Baseline Models*: To evaluate the performance of our proposed FOYF, we compare it against seven representative methods spanning both standard FSS and cutting-edge CD-FSS. They include PFENet (prototype-based) [6] and HSNet (affinity-based) [19], which are classical FSS approaches relying on convolutional feature patterns. For CD-FSS, we evaluate state-of-the-art models: PATNet [4], ABCDFSS [30], DMTNet [28], IFA [29], and PFMNet [3]. All method implementations utilize publicly available code and default hyperparameters to ensure fair comparison.

3) *Training and Testing Strategies*: Metatraining is conducted across all PASCAL VOC classes with cross-entropy and Dice losses [37], while metatesting is performed independently on each target dataset. Performance is evaluated using mean intersection-over-union (mIoU) and foreground-background intersection-over-union (FB-IoU) [4], averaged over five runs. Each experimental run includes 200 tasks across all datasets, with the exception of FSS-1000, which employs 1000 tasks per run to accommodate its larger dataset scale.

B. Evaluation Metrics

We report two metrics: mIoU and FB-IoU. For a given class, $\text{IoU} = \text{TP}/(\text{TP} + \text{FP} + \text{FN})$, where TP, FP, and FN denote true positive, false positive, and false negative pixel counts in the predicted segmentation mask. These metrics provide a comprehensive assessment of both classwise and overall segmentation quality.

C. Implementation Details

FOYF employs a Swin Transformer as the backbone for self-attention pattern learning, initialized with weights pre-trained on ImageNet. To balance computational efficiency and performance, input images (both support and query) are resized to 224×224 pixels during training. The model is implemented in PyTorch and optimized using the Adam optimizer with a learning rate of $1e^{-4}$, a total of 300 iterations, and a batch size of 4. During inference, images are similarly resized to 224×224 to maintain consistency. All experimental training and testing procedures are executed on an NVIDIA GeForce RTX 4090 GPU with 24 GB of memory.

A critical modification to prior protocols is given as follows. First, categories within each benchmark dataset are classified, and the number of images under each category is counted simultaneously. Subsequently, the images in each category are equally divided, with half being used to construct the validation set and the other half for constructing the test set. This replaces the conventional practice of reusing the entire dataset for both validation and testing, ensuring a more reliable performance estimate. In addition, all models are evaluated solely on their end-to-end outputs, excluding postprocessing steps, to isolate the true impact of architectural innovations.

D. Performance Analysis

Table I presents a quantitative evaluation of various state-of-the-art approaches on the challenging CD-FSS benchmark under a five-shot setting, focusing on the mIoU and FB-IoU metrics. Classical FSS methods, such as PFENet and HSNet, while effective in standard scenarios, exhibit significant performance drops when confronted with substantial domain shifts between source and target domains. This limitation becomes particularly pronounced on datasets characterized by large domain gaps, such as Chest X-ray (medical imaging), ISIC2018 (dermatology), and Deepglobe (satellite imagery). The core weakness of these classical approaches stems from their reliance on local feature pattern exploitation. Class-level prototype strategy of PFENet and 4-D kernel affinity mechanism of HSNet, though innovative, fundamentally depend on capturing intraclass similarities and interclass differences through convolutional operations. However, the representative power of such local feature patterns proves inadequate for bridging substantial domain disparities, hindering effective model generalization. In stark contrast, our proposed FOYF model demonstrates robust cross-domain performance by explicitly addressing the domain gap through attention specialization. This advantage is quantitatively evident across the three challenging datasets. Compared to PFENet and HSNet, FOYF achieves substantial mIoU improvements by 31.22% and 12.64% on Chest X-ray, by 24.90% and 9.89% on ISIC2018, and by 11.42% and 6.57% on Deepglobe. While FOYF achieves comparable performance to HSNet in the FSS-1000 dataset (where the domain gap is smaller), its superiority in cross-domain settings is clear. Notably, in terms of the mIoU of data, FOYF secures the top position under the five-shot setting for both ISIC2018 and Deepglobe datasets, surpassing the closest competitors by significant margins of 2.96% on

TABLE I
mIoU (%) AND FB-IoU (%) RESULTS OF VARIOUS METHODS IN ONE-WAY FIVE-SHOT ON CD-FSS BENCHMARK

Methods Metrics	FSS-1000		Chest X-ray		ISIC2018		Deepglobe		Average		Time(s)
	mIoU	FB-IoU	mIoU	FB-IoU	mIoU	FB-IoU	mIoU	FB-IoU	mIoU	FB-IoU	
PFENet [6]	71.07	79.59	37.03	51.20	29.27	34.91	40.02	38.89	44.35	51.15	0.045
HSNet [19]	73.86	83.21	55.61	64.54	44.28	58.01	44.87	41.98	54.66	61.94	0.059
PATNet [4]	72.29	82.04	66.00	75.28	48.86	62.69	41.58	44.30	57.18	66.08	0.148
ABCDFFS [30]	68.07	78.55	73.45	80.78	51.21	59.68	48.14	50.54	60.22	67.39	0.062
DMTNet [28]	70.51	80.80	41.03	59.92	43.42	57.87	43.51	44.05	49.62	60.66	0.714
IFA [29]	65.12	76.65	71.53	80.81	43.91	62.07	45.18	46.16	56.44	66.42	0.069
PFMNet [3]	70.04	80.24	54.03	64.35	35.38	44.78	46.63	28.57	51.52	54.49	0.030
FOYF(Ours)	73.77	82.21	68.25	77.23	54.17	63.52	51.44	53.36	61.91	69.08	0.056

Note that the **bold** data represents the best, and the **bold italic** data represents the second best.

ISIC2018 and 3.3% on Deepglobe, respectively. On the Chest X-ray dataset, FOYF attains the third-highest performance, scoring 68.25% mIoU and 77.23% FB-IoU under the five-shot setting. Averaging across all evaluated datasets, FOYF consistently delivers the best overall performance, achieving an average mIoU of 61.91% and an average FB-IoU of 69.08%. In addition, it also presents the runtime of different methods. As shown in the data, the proposed FOYF method has relatively low computational time, ranking third only after PFMNet and PFENet, but its cross-domain segmentation performance significantly outperforms them. The results in Table I conclusively demonstrate the effectiveness of FOYF in tackling the demanding CD-FSS tasks, particularly when significant domain gaps exist.

E. Visualization Analysis

Fig. 3 visualizes the segmentation performance of various models on the CDFSS benchmark during inference. To facilitate clear comparison, the query images are annotated with color-coded regions: green denotes regions correctly identified as the target class (true positive), blue indicates regions incorrectly marked as the target class (false positive), and red highlights regions incorrectly missed (false negative). Through observing all the visual comparison results, we find that models such as PFENet, HSNet, and PFMNet [e.g., rows in Fig. 3(b), (c), and (h)] exhibit substantial false positive regions when operating in cross-domain scenarios. The inherent convolutional inductive bias of these models restricts their ability to capture scene-level context, making it difficult to distinguish foreground from background, which, in turn, exacerbates the false-positive issue. Consequently, these methods struggle to reliably distinguish objects of interest from the surrounding background or distractor elements in unfamiliar domains. Besides, PATNet and DMTNet [e.g., Fig. 3(d5)–(d8) and (f3)–(f7)] demonstrate segmentation lacking completeness, characterized by fragmented regions and noticeable missing areas. Their shared strategy of transforming domain-specific features into domain-agnostic representations, while mitigating some cross-domain challenges, inherently discards critical scene-level semantic information vital for maintaining the structural integrity and coherence of the segmented objects. Methods incorporating fine-tuning, namely, ABCDFSS and IFA [e.g., rows in Fig. 3(e) and (g)], show improvement over the aforementioned models, exhibiting fewer false positive and false negative

TABLE II
SEGMENTATION IMPACT (%) OF SAPLM, GAPLM, ASFM, DSFP, AND MAP FOR THE CD-FSS BENCHMARK (FSS-1000 AND CHEST X-RAY)

Methods Metrics	FSS-1000		Chest X-ray		Average	
	mIoU	FB-IoU	mIoU	FB-IoU	mIoU	FB-IoU
w ALL	73.77	82.21	68.25	77.23	71.01	79.72
w/o SAPLM	46.67	64.37	40.82	53.53	43.75	58.95
w/o GAPLM	49.65	65.90	53.34	63.52	51.50	64.71
w/o ASFM	60.34	72.84	38.58	54.04	49.46	63.44
w/o DSFP	49.41	63.88	42.35	54.50	45.88	59.19
w/o MAP	70.10	79.13	65.33	72.58	67.72	75.86
w SAPLM	48.95	60.82	38.06	53.22	43.51	57.02

Note that ‘w’ or ‘w/o’ denotes FOYF model with or without a specific module. ‘ALL’ represents the original version of FOYF.

regions. However, the better performance is observed with our proposed FOYF model. Fig. 3(i1)–(i8) shows that FOYF delivers superior segmentation robustness and accuracy to competing methods, though its performance still exhibits room for refinement. This superiority arises from its attention specialization, which is designed to mine and fuse scene-level context under few-shot constraints. By balancing holistic scene comprehension with local detail preservation, FOYF achieves an equilibrium between segmentation precision and completeness, underscoring its strong competitive edge in CD-FSS tasks. However, boundary detection weaknesses persist, stemming primarily from patch-merging feature compression that erodes fine details. Although the original image is incorporated in multiscale attention perception classification to supplement detailed information, its efficacy demands further enhancement for robust boundary delineation.

F. Ablation Study

This section presents a comprehensive ablation study to isolate and evaluate the individual contributions of attention specialization modules of FOYF under varying domain conditions. To achieve this, we strategically select FSS-1000 with a small domain gap and Chest X-ray with a large domain gap compared with Pascal VOC 2012. Table II details the five-shot ablation results, focusing on the impact of five key components: the SAPLM, the guide-attention pattern learning module (GAPLM), the ASFM, the deep semantic feature patterns (DSFPs), and the MAP of FOYF. The quantitative analysis reveals their distinct roles in enhancing segmentation performance (mIoU and FB-IoU). The foundational improvement comes from the SAPLM, which establishes

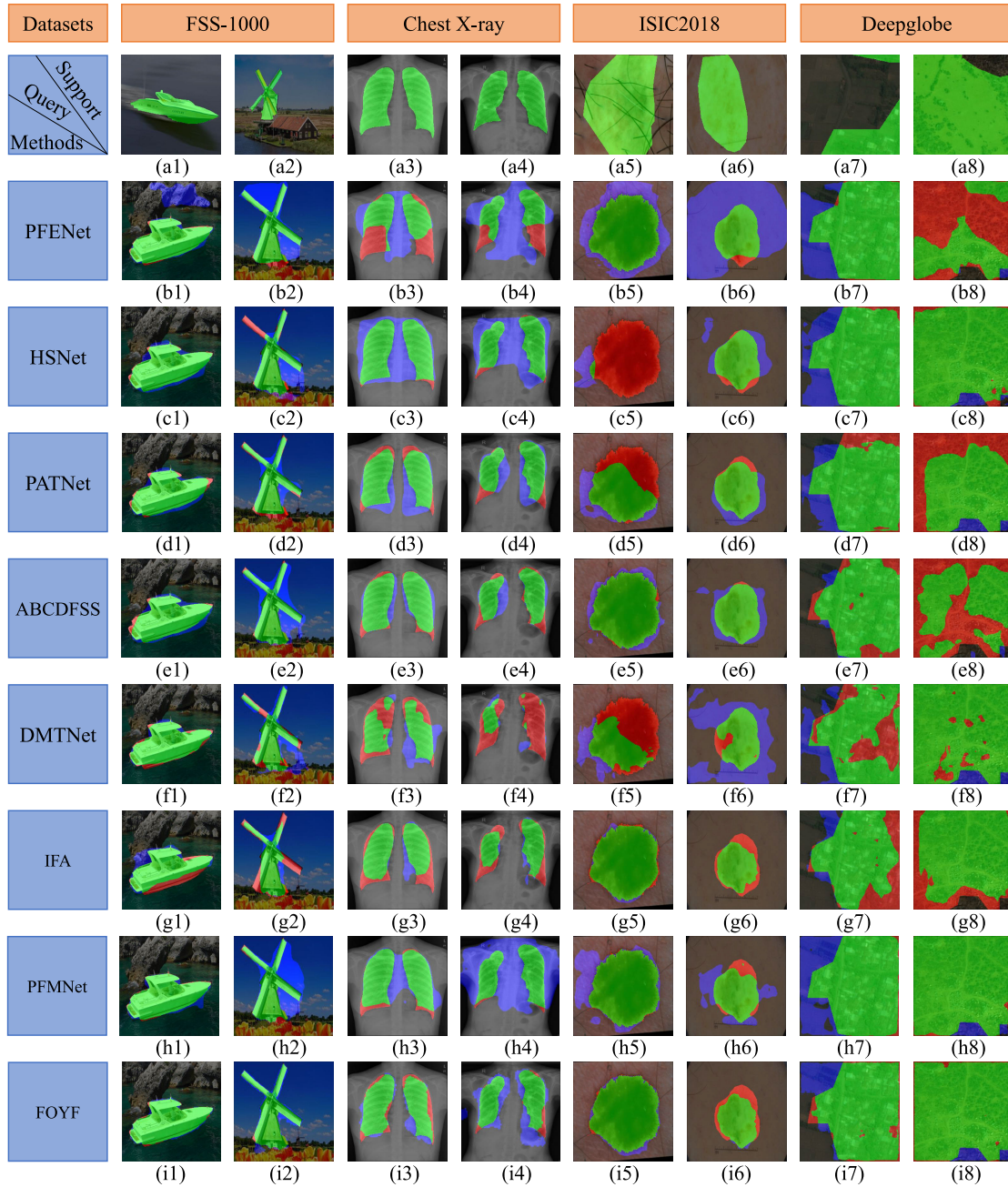


Fig. 3. Visual comparison of segmentation results produced by different methods in five-shot settings for CDFSS benchmark, i.e., FSS-1000 (columns 1 and 2), Chest X-ray (columns 3 and 4), ISIC2018 (columns 5 and 6), and Deepglobe (columns 7 and 8). Images from top to bottom are support image samples in the target domain with ground truth in green [row (a)] and inference results of query images in the target domain by PFENet, HSNet, PATNet, ABCDFSS, DMTNet, IFA, PFMNet, and FOYF [rows (b)–(i)]. True positive, false positive, and false negative of inference results are labeled in green, blue, and red colors in the query images, respectively.

a strong baseline by enabling the model to capture self-correlation patterns. Compared to the baseline model with all attention specialization modules, SAPLM alone provides a substantial boost of 27.26% in average mIoU and 20.77% in average FB-IoU. Building upon this foundation, GAPLM further refines the learned patterns by incorporating external guidance mechanisms. This results in an additional significant gain of 19.51% in average mIoU and 15.01% in average FB-IoU. ASFM then contributes by effectively integrating the diverse attention patterns learned by the previous modules. Its

inclusion yields improvements of 21.55% in average mIoU and 16.28% in average FB-IoU compared to the baseline without ASFM. DSFP ensures a balanced enhancement across both metrics, delivering 25.13% in average mIoU and 20.53% in average FB-IoU gains relative to the baseline without DSFP. Finally, the MAP module with multiple window sizes, relative to a single-scale attention perceptron module, contributes 3.29% mIoU and 3.86% FB-IoU to the baseline on average, because multiple window sizes help capture multiscale context. Notably, the relative contribution of these modules

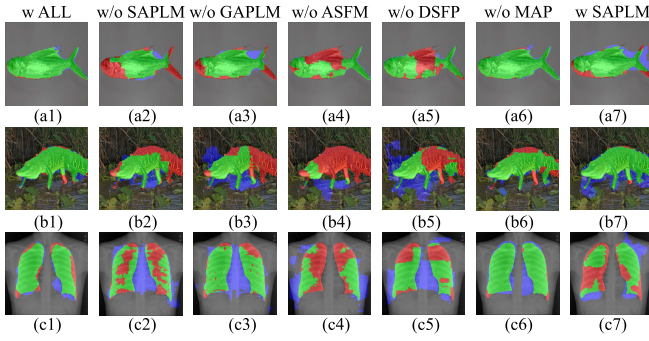


Fig. 4. Ablation study of FOYF, containing original version (w ALL), version without SAPLM (w/o SAPLM), version without GAPLM (w/o GAPLM), version without ASFM (w/o ASFM), version without DSFP (w/o DSFP), version without MAP (w/o MAP), and version only with SAPLM (w SAPLM). The query results of these versions are illustrated in images with clean backgrounds (a1)–(a7), complex scenes (b1)–(b7), and a large domain gap (c1)–(c7). True positive, false positive, and false negative results in query images are labeled in green, blue, and red.

varies slightly depending on the inherent domain gap of the data. Comparing the performance metrics on Chest X-ray and FSS-1000 reveals that SAPLM, ASFM, GAPLM, and DSFP provide relatively larger benefits for CD-FSS compared to MAP.

Fig. 4 visually demonstrates the impact of each attention specialization module on the quality of segmentation results. To comprehensively assess performance under varying conditions, we select query samples representing distinct challenges: a flying fish class image with clean backgrounds (row 1), a short crocodile class image with complex scenes (row 2), and the CHNCXR class image from the Chest X-ray dataset (row 3) with a large domain gap. True positive, false positive, and false negative regions are color-coded in green, blue, and red, respectively, within the query images, enabling intuitive comparison. As depicted in the second column of Fig. 4, excluding the SAPLM leads to the loss of feature pattern regions sharing similar attributes. This deficiency is particularly pronounced in the challenging Chest X-ray dataset [see Fig. 4(c2)], highlighting SAPLM’s crucial role in capturing essential self-correlation patterns, especially under large domain gaps. The segmentation results in Fig. 4(b3) and (c3) reveal a significant increase in false positive regions when the GAPLM is omitted. This indicates that without GAPLM’s guidance, the model struggles to effectively suppress irrelevant scene-level feature patterns previously mined by SAPLM, leading to oversegmentation. Replacing the ASFM with hard fusion results in a noticeable degradation in segmentation quality across all datasets, as shown in Fig. 4(a4)–(c4). This decline stems from hard fusion ignoring the diaphragmatic response of feature patterns across different stages, demonstrating ASFM’s essential role in adaptively integrating hierarchical scene-level information. The fragmentation observed in the “w/o DSFP” underscores the importance of deep-level feature patterns [see Fig. 4(a5)–(c5)]. Relying solely on shallow-level features hampers the model’s ability to discern relationships between objects and their context within the scene, resulting in incomplete and disconnected segmentation. It emphasizes DSFP’s contribution to capturing high-level information vital

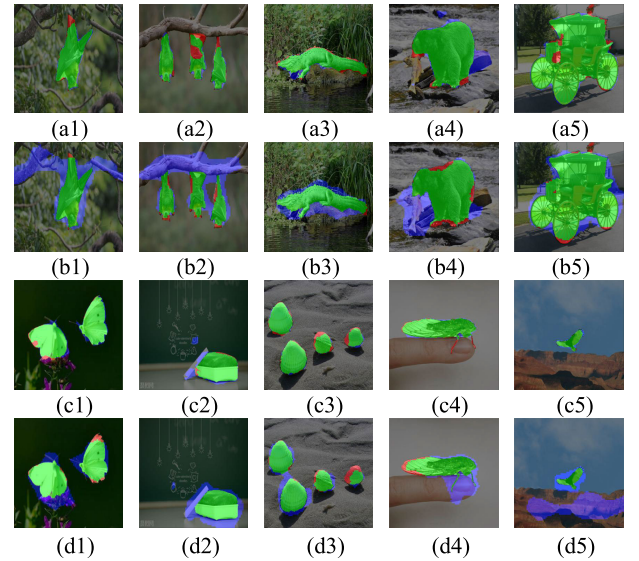


Fig. 5. Impact of modeling foreground-background relationships before masking on few-shot segmentation tasks with complex backgrounds and a small foreground proportion: results in row (a) are modeling before masking operation on samples with complex backgrounds; results in row (b) are a direct masking operation on samples with complex backgrounds; results in row (c) are modeling before the masking operation on samples with a small foreground proportion; and results in row (d) are a direct masking operation on samples with a small foreground proportion.

for CD-FSS. Fig. 4(b6) and (c6) demonstrate that the single-scale attention perceptron module with the default window size of 7 in the Swin Transformer block (i.e., “w/o MAP”) exhibits poor segmentation accuracy in complex and cross-domain scenarios. In contrast, the baseline equipped with an MAP module, which incorporates three different window sizes of 3, 5, and 7, simultaneously captures multiscale contextual information, enabling complementary information fusion and, thus, achieving more precise segmentation of target boundaries.

By quantitatively and visually analyzing the progressive deterioration in segmentation quality as modules are removed, the critical and interdependent roles of SAPLM, GAPLM, ASFM, DSFP, and MAP become evident. The single-attention approach, like the version solely employing SAPLM (i.e., “w SAPLM” in Table II and Fig. 4), is experimentally proven to be inadequate to cope with the complexity across different scenarios. It fails to focus its attention on the target. On the contrary, the combined use of multiple attention-based modules yields a synergistic solution that effectively addresses the complexities of the CD-FSS task. Each one addresses a distinct challenge to CD-FSS: SAPLM captures fundamental patterns, GAPLM refines them with guidance, ASFM fuses them adaptively, DSFP provides deep scene-level context, and MAP refines boundary segmentation.

V. DISCUSSION

Equation (1) first models the foreground-background relationships and then filters out the mask background part, which can preserve more scene-level information. To validate this viewpoint, we meticulously selected samples from the

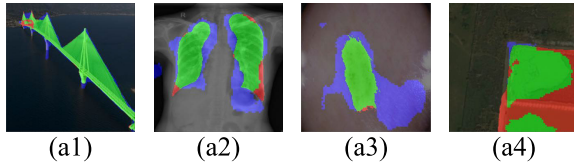


Fig. 6. Failure modes. (a1)–(a4) Some failure cases on CD-FSS benchmark.

FSS-1000 dataset for experimental purposes. Specifically, we chose 20 samples with complex backgrounds and 20 samples where the foreground accounted for approximately 1/16 of the total image area. They are used to test the impact of two operations: modeling the foreground–background relationship before masking and directly masking the background on the few-shot segmentation task. As shown in Fig. 5, in the few-shot segmentation task, the performance of modeling the foreground–background relationship before masking significantly outperforms that of directly masking the background. The former enables a more complete segmentation result, while the latter often introduces more false-positive regions (marked in blue in the figure). For samples with complex backgrounds, the mIoU obtained through the direct masking operation is 0.52. In contrast, the mIoU achieved by modeling the foreground–background relationship before masking is 0.83, representing a substantial performance gain of up to 59.48%. For samples with a relatively small foreground proportion, the mIoU for direct masking is 0.48, whereas the mIoU for modeling the foreground–background relationship before masking reaches 0.70, yielding a gain of 47.03%. Based on the above experimental results, it can be convincingly demonstrated that modeling the foreground–background relationship before masking enables the retention of more scene-level information.

In addition, we analyze the failure modes in the benchmark tests, as illustrated in Fig. 6. For natural-scene images [see Fig. 6(a1)], the support image only encompasses the upper part of the bridge, whereas the actual model prediction also includes the bridge piers. The primary reason for this discrepancy lies in the fact that our proposed method differs from previous methods. They only interact with shallow-level information; in contrast, our method not only interacts with shallow-level information but also fully leverages semantic information, enabling it to grasp the holistic nature of the target. For medical images [refer to Fig. 6(a2) and (a3)], when shadows appear in the neighborhood background of the target, this can adversely affect the model’s segmentation performance, resulting in the model’s predictions containing a few false-positive regions. This is mainly because the PASCAL VOC 2012 dataset used during model training does not contain such latent patterns, leaving the model without corresponding learning samples. In the case of remote sensing images [refer to Fig. 6(a4)], significant appearance differences between similar targets in the support and query images lead the model to easily miss detecting certain regions. In light of the above findings, we can further research and optimize the model in subsequent studies.

VI. CONCLUSION

This article proposes a novel network, FOYF, for CD-FSS. Multiple attention specialization modules based on Swin Transformer are designed to solve the key challenges of feature representation and cross-domain propagation in CD-FSS. FOYF introduces four collaborative innovations. SAPLM adaptively mines internal attributes of the target in the scene, GAPLM dynamically activates cross-scene associations, ASFM fuses hierarchical feature patterns in the scene, and the multiscale attention perception classification module optimizes query segmentation via scale-aware attention. Experimental results across cross-domain benchmarks show that FOYF has significantly improved accuracy and robustness, reaching advanced performance. These advancements stem from its core innovation, attention specialization, which explicitly prioritizes holistic scene understanding while preserving domain-invariant local patterns. By transcending the limitations of traditional convolutional bottlenecks, FOYF establishes a new method for efficient CD-FSS, offering new insights for algorithm optimization. Future work will investigate edge-aware loss integration and multiscale feature fusion in CD-FSS to improve segmentation accuracy.

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